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Sec: 11

Assignment: 3

CSE-423.

Ans-01

For RGB (255, 165, 0) $\rightarrow$ HSV:

$$\text{Normalize: } r = \frac{255}{255} \quad g = \frac{165}{255} \quad b = \frac{0}{255}$$
$$= 1 \quad = 0.647 \quad = 0$$

here, $c_{\max} = 1$ $c_{\min} = 0$.

$$\text{diff} = 1 - 0 = 1$$

hue:

$$h = \left(\frac{g-b}{\text{diff}} \right) \times 60$$

$$= \frac{0.647}{1} \times 60 = 38.82^\circ$$

saturation:

$$\frac{\text{diff}}{c_{\max}} \times 100 = \frac{1}{1} \times 100$$

$$= 100\%$$

Value:

$$c_{\max} \times 100 = 1 \times 100 = 100\%$$

$$\text{HSV} = (38.82, 100, 100)$$

By

For RGB(255, 105, 160) $\rightarrow$ HSV:

$$r = 1 \quad g = 0.412 \quad b = 0.706$$

$$C_{\max} = 1(R) \quad C_{\min} = 0.412(G)$$

$$\text{diff} = 0.588$$

$$\text{Hue: } h = 60 \times \left(\frac{g - b}{\text{diff}} \right)$$

$$= 60 \times \left(\frac{0.412 - 0.706}{0.588} \right)$$

$$= 60 \times \left(\frac{-0.294}{0.588} \right) = -30 + 360^\circ$$

$$= -30^\circ = 330^\circ$$

Saturation:

$$S = \frac{\text{diff}}{C_{\max}} = \frac{0.588}{1}$$

$$= 0.588 \times 100$$

$$= 58.8\%$$

Value:

$$V = C_{\max}$$

$$= 1 \times 100 = 100\%$$

$$\text{HSV} = (330, 0.588, 1)$$

Ans to the Q/No-02

HSV $\rightarrow$ RGB

For $(38.8, 1, 1) \rightarrow \text{RGB}:$

$$H = 38.8 \quad S = 1 \quad V = 1$$

$$C = V \times S = 1$$

$$X = C \times \left(\frac{1}{3} - \left| \left(\frac{H}{60} \bmod 2 \right) - 1 \right| \right)$$

$$= 1 \times \left(\frac{1}{3} - \left| \left(\frac{38.8}{60} \bmod 2 \right) - 1 \right| \right)$$

$$= 1 - \left| 0.64 \bmod 2 - 1 \right|$$

$$= 1 - \left| 0.64 - 1 \right|$$

$$= 1 - 0.36$$

$$= 0.64$$

$$m = V - C = 1 - 1 = 0$$

$$0 \leq H < 60; \quad R = C, \quad G = X, \quad B = 0$$

$$R = 1, \quad G = 0.64, \quad B = 0$$

Adjustment $(R, G, B) + (m, m, m); (R, G, B) = (1, 0.64, 0)$

Scale, $(R, G, B) = (255, 165, 0)$

$$\text{HSV}(330, 0.588, 1) \rightarrow \text{RGB:}$$

$$\begin{aligned} C &= V \times S. & \left| \begin{array}{l} V=1 \\ S=0.588. \end{array} \right. \\ &= 1 \times 0.588 \\ &= 0.588. \end{aligned}$$

$$X = C \times \left(1 - \left| \left(H/60 \bmod 2 \right) - 1 \right| \right)$$

$$= 0.588 \left(1 - \left| \left(330/60 \bmod 2 \right) - 1 \right| \right)$$

$$= 0.588 \left(1 - \left| \left(5.5 \bmod 2 \right) - 1 \right| \right)$$

$$= 0.588 \left(1 - \left| 0.5 \right| \right)$$

$$= 0.294$$

$$m = V - C = 1 - 0.588 = 0.412$$

$$\begin{aligned} 300 \leq H < 360^\circ; & \quad R = C \quad G = 0 \quad B = X \\ & \quad = 0.588 \quad G = 0 \quad B = 0.294. \end{aligned}$$

$$\begin{aligned} \text{Adjustment: } R' &= R + m & G' &= G + m & B' &= B + m \\ &= 1 & &= 0.412 & &= 0.706. \end{aligned}$$

$$\text{scaling, RGB} = (255, 105, 180)$$

Ans to Q/No-3

For RGB(255, 165, 0) $\rightarrow$ HLS:

Normalise, $r = 1$ $g = 0.647$ $b = 0$.

$$L = \frac{C}{2} = \frac{C_{\max} + C_{\min}}{2} = \frac{1}{2} = 0.5$$

As, $L \leq 0.5$;

$$S = \frac{d}{C} = \frac{d}{C_{\max} - C_{\min}} = \frac{1}{1} = 1$$

$C_{\max}$ for r, g, b

$$S_o, H_o = (g - b) / d = (0.647 / 1) 60^\circ$$

$$H_o = 0.647 \times 60^\circ$$

$$(inverted) = 38.82^\circ$$

$$HLS = (38.82, 0.5, 1)$$

Ans

For, RGB(255, 185, 180) $\rightarrow$ HLS:

$$r = 1 \quad g = 0.412 \quad b = 0.706$$

$$L = (1 + 0.412) / 2 \\ = 0.706$$

As, $L > 0.5$

~~$$S = 2 - \left(\frac{d}{C_m + C_m} \right)$$~~

~~$$= 2 - \left(\frac{1 - 0.412}{1 + 0.412} \right)$$~~

$$S = \frac{\text{diff}}{2 - (C_m + C_{mi})}$$

$$= \frac{1 - 0.412}{2 - (1 - 0.412)} = \frac{0.588}{0.588}$$

$$H = \left((g - b) / d \right) \times 60$$

$$= -30^\circ + 360^\circ$$

$$= 330^\circ$$

Ans: HSL = (330°, 1, 0.706)